

Pattern Program: Java

```
import java.util.Scanner;
public class Lab1_Pattern {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        Lab1_Pattern lp=new Lab1_Pattern();
        int n=5;
        lp.ptrn1(n);
        lp.ptrn2(n);
        lp.ptrn3(n);
        lp.ptrn4(n);
        lp.ptrn5(n);
        lp.ptrn6(n);
        lp.ptrn7(n);
        lp.ptrn8(n);
        lp.ptrn9(n);
        lp.ptrn10(n);
        lp.ptrn11(n);
        lp.ptrn12("Darshan");
    } //main

    void ptrn1(int n){
        System.out.println("Square:");
        for(int i=0;i<n;i++){
            for(int j=0;j<n;j++){
                System.out.print("*");
            }
            System.out.println();
        }
    }

    void ptrn2(int n){
        System.out.println("Left Triangle");
        for(int i=0;i<n;i++){
            for(int j=0;j<=i;j++){
                System.out.print("*");
            }
            System.out.println();
        }
    }

    void ptrn3(int n){
        //left triangle reverse
        System.out.println("left triangle rev");
        for(int i=0;i<n;i++){
            for(int j=0;j<n-i;j++){
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Square:

```
*****
*****
*****
*****
*****
```

Left
Triangle

```
*
**
***
****
*****
```

left
triangle
reverse

```
*****
****
***
**
*
```

Pattern Program: Java

```
void ptrn4(int n){
    System.out.println("right triangle");
    for(int i=0;i<n;i++){
        for(int j=0;j<n-i-1;j++){
            System.out.print(" ");
        }
        for(int k=0;k<=i;k++){
            System.out.print("*");
        }
        System.out.println();
    }
}
```

```
right
triangle
*
**
***
****
*****
```

```
void ptrn5(int n){
    System.out.println("right triangle reverse");
    for(int i=0;i<n;i++){
        for(int j=0;j<=i-1;j++){
            System.out.print(" ");
        }
        for(int k=n-1;k>=i;k--){
            System.out.print("*");
        }
        System.out.println();
    }
}
```

```
right
triangle
reverse
*****
****
***
**
*
```

```
void ptrn6(int n){
    System.out.println("Hollow Square");
    for(int i=0;i<n;i++){
        for(int j=0;j<n;j++){
            if(j==0||j==n-1||i==0||i==n-1)
                System.out.print("*");
            else
                System.out.print(" ");
        }
        System.out.println();
    }
}
```

```
Hollow
Square
*****
*   *
*   *
*   *
*****
```

```
void ptrn7(int n){
    System.out.println("middle triangle");
    for(int i=0;i<n;i++){
        for(int j=0;j<n-i-1;j++){
            System.out.print(" ");
        }
        for(int k=0;k<=i;k++){
            System.out.print("* ");
        }
        System.out.println();
    }
}
```

```
middle
triangle
*
* *
* * *
* * * *
* * * * *
```

Pattern Program: Java

```
void ptrn8(int n){
    System.out.println("Diamond");
    for(int i=0;i<n-1;i++){
        for(int j=0;j<n-i-1;j++){
            System.out.print(" ");
        }
        for(int k=0;k<=i;k++){
            System.out.print("* ");
        }
        System.out.println();
    }
    //rev
    for(int i=0;i<n;i++){
        for(int j=0;j<i;j++){
            System.out.print(" ");
        }
        for(int k=n-1;k>=i;k--){
            System.out.print("* ");
        }
        System.out.println();
    }
}
```

Diamond

```

      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * *
* * * * *
 * * *
  * *
   *
```

```
void ptrn9(int n){
    System.out.println("Hollow Diamond");
    for(int i=0;i<n-1;i++){
        for(int j=0;j<n-i-1;j++){
            System.out.print(" ");
        }
        for(int k=0;k<=i;k++){
            if(k==n || k==i || k==0 || k==n-1)
                System.out.print("* ");
            else
                System.out.print(" ");
        }
        System.out.println();
    }
    //rev
    for(int i=0;i<n;i++){
        for(int j=0;j<i;j++){
            System.out.print(" ");
        }
        for(int k=n-1;k>=i;k--){
            if(k==i || k==0 || k==n-1)
                System.out.print("* ");
            else
                System.out.print(" ");
        }
        System.out.println();
    }
}
```

Hollow
Diamond

```

      *
     * *
    *   *
   *     *
  *       *
 *         *
*         *
 *       *
  *     *
   *   *
    * *
     *
```

Pattern Program: Java

```
void ptrn10(int n){
    for(int i=1;i<=n;i++){
        for(int j=1;j<=i;j++){
            System.out.print(i);
        }
        System.out.println();
    }
}

void ptrn11(int n){
    for(int i=1;i<=n;i++){
        for(int j=1;j<=i;j++){
            System.out.print(j);
        }
        System.out.println();
    }
}

void ptrn12(String s){
    for(int i=0;i<s.length();i++){
        for(int j=0;j<=i;j++){
            System.out.print(s.charAt(j));
        }
        System.out.println();
    }
}
```

1
22
333
4444
55555

1
12
123
1234
12345

D
Da
Dar
Dars
Darsh
Darsha
Darshan